

Applied Physics Lab Handbook

Complete professional laboratory textbook for Course Code 2006. This lesson is built from the supplied official syllabus and converts the experiment list into a practical learning, revision, viva and exam-preparation handbook.

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COURSE CODE

2006

TITLE

Applied Physics Lab

SEMESTER

1 & 2

CREDITS

1

CATEGORY

Basic Science

PERIODS

2/week

SEMESTER PERIODS

30 + 30

SCHEME

Revision 2021

[Engineering lab flow](#)

Measure → Correct → Calculate → Verify → Report

Every practical answer must show aim, apparatus, principle, formula, diagram/circuit, procedure, observation table, calculation, result, precautions and common errors.

20 experiments

4 CO modules

12 minimum experiments to perform



Objectives from supplied syllabus

- Supplement factual physics knowledge by first-hand manipulation of apparatus.
- Develop scientific temper for applying physics principles to engineering and technology problems.
- Build confidence in equipment handling and measurement skills.

Prerequisite

Basic knowledge in Physics from Secondary School.

ഈ lab subject-ൽ theory മാത്രം മതിയാകില്ല. Instrument വായിക്കാൻ, least count കണ്ടെത്താൻ, graph വരയ്ക്കാൻ, result unit സഹിതം എഴുതാൻ practice വേണം.

Table of Contents

Official source information

Item	Status	Value from supplied syllabus/source
Course code/title	Specified	2006 - Applied Physics Lab
Programme	Specified	Diploma in Engineering and Technology
Semester	Specified	1 & 2
Credits/category	Specified	1 credit, Basic Science
Periods	Specified	2 per week; 30 + 30 per semester
CO-PO mapping	Specified	CO1, CO2, CO3 and CO4 mapped to PO1 at level 3. Other PO values are blank/not specified.
Exam blueprint	Limited	Lab Exam I and II are mentioned; detailed written question-paper pattern is not specified in supplied source. Model paper here is a fresh practice paper.

Redesigned Syllabus Roadmap

Not a copied syllabus table. This roadmap tells what to draw, calculate, explain and avoid in each module.

Module 1 • 12 hours • CO1**Measurement Tools and Accuracy**

Outcome: Select appropriate measuring tools and make measurements with accuracy and precision.

Core topics: Vernier calipers, Screw gauge, Spherometer, Mercury thermometer, Least count, Zero error, Unit conversion

What to draw: Vernier jaws and scales, Screw gauge pitch/head scale, Spherometer legs and screw, Temperature scale conversion

What to calculate: least count, zero correction, volume, diameter/thickness, radius of curvature, temperature conversion

Field relevance: Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

Module 2 • 23 hours • CO2**Mechanics and Properties of Matter**

Outcome: Apply and illustrate mechanics and properties of matter through experiments.

Core topics: Hooke law, Stoke law, Parallelogram law, Moment bar, U-tube, Flywheel, Pendulum, Resonance column, Cantilever

What to draw: spring graph, terminal velocity setup, vector force diagram, moment bar, U-tube levels, flywheel, pendulum, resonance column, cantilever

What to calculate: spring constant, viscosity, unknown mass, moment balance, relative density, moment of inertia, g, sound velocity, time period

Field relevance: Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

Module 3 • 6 hours • CO3**Ray Optics Experiments**

Outcome: Experiment with lens, prism and glass slab to realize basic laws of ray optics.

Core topics: Convex lens u-v method, Lens power, Refraction, Snell law, Ray tracing

What to draw: u-v setup, convex lens ray diagram, glass slab/prism ray diagram

What to calculate: focal length, power, $\sin i / \sin r$, refractive index

Field relevance: Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

Module 4 • 15 hours • CO4

Electrical and Semiconductor Characteristics

Course 2006

Outcome: Use V-I characteristics of conductors and semiconductors to determine resistance of materials.

Core topics: Ohm law, series/parallel resistance, diode V-I, potentiometer, meter bridge, specific resistance

What to draw: ammeter-voltmeter circuit, series/parallel circuits, diode V-I graph, potentiometer circuit, meter bridge

What to calculate: $R=V/I$, equivalent resistance, knee voltage, internal resistance, specific resistance

Field relevance: Reading accuracy, zero correction, graph plotting, connection checking and troubleshooting are the same habits needed in workshop testing, electrical panels, sensors and maintenance measurements.

Formula / Rule Bank

Formula, variable meaning, unit and use. For lab exams, write formula before substitution and final result with unit.

Vernier LC

$$LC = 1 \text{ MSD} - 1 \text{ VSD}$$

Unit: mm/cm

Use: smallest Vernier reading

Example: Substitute readings with units and write final result correctly.

Screw gauge LC

$$LC = \text{pitch} / \text{circular divisions}$$

Unit: mm

Use: fine thickness/diameter

Example: Substitute readings with units and write final result correctly.

Sphere volume

$$V = \frac{4}{3} \pi r^3$$

Unit: cm^3

Use: sphere volume

Example: Substitute readings with units and write final result correctly.

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Cylinder volume

$$V = \pi r^2 h$$

Unit: cm^3

Use: cylinder/test tube

Example: Substitute readings with units and write final result correctly.

Spherometer

$$R = \frac{l^2}{6h} + \frac{h}{2}$$

Unit: cm

Use: radius of curvature

Example: Substitute readings with units and write final result correctly.

Temperature

$$K = C + 273; F = \frac{9C}{5} + 32$$

Unit: K/F

Use: scale conversion

Example: Substitute readings with units and write final result correctly.

Hooke law

$$F = kx$$

Unit: N/m

Use: spring constant

Example: Substitute readings with units and write final result correctly.

Course: DP205

$$\eta = \frac{2r^2(\rho_{\text{sol}} - \rho_{\text{liq}})g}{9v}$$

Unit: Pa s

Use: viscosity

Example: Substitute readings with units and write final result correctly.

Moment

$$W_1d_1 = W_2d_2$$

Unit: N m

Use: moment bar

Example: Substitute readings with units and write final result correctly.

Pendulum

$$g = \frac{4\pi^2 l}{T^2}$$

Unit: m/s²

Use: gravity

Example: Substitute readings with units and write final result correctly.

Sound

$$v = 2f(l_2 - l_1)$$

Unit: m/s

Use: resonance column

Example: Substitute readings with units and write final result correctly.

Lens

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Unit: m

Use: focal length

Example: Substitute readings with units and write final result correctly.

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Lens power

$$P=1/f$$

Unit: D

Use: lens power

Example: Substitute readings with units and write final result correctly.

Snell

$$\mu=\sin i/\sin r$$

Unit: none

Use: refractive index

Example: Substitute readings with units and write final result correctly.

Ohm

$$V=IR$$

Unit: ohm

Use: resistance

Example: Substitute readings with units and write final result correctly.

Series R

$$R_s=R_1+R_2+\dots$$

Unit: ohm

Use: series circuit

Example: Substitute readings with units and write final result correctly.

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

Unit: ohm

Use: parallel circuit

Example: Substitute readings with units and write final result correctly.

Diode

$$R_f = \Delta V / \Delta I$$

Unit: ohm

Use: V-I curve

Example: Substitute readings with units and write final result correctly.

Meter bridge

$$\rho = RA/L$$

Unit: ohm m

Use: specific resistance

Example: Substitute readings with units and write final result correctly.

Practice, Expected Questions and Quick Revision

Module-wise viva, short-answer, numerical, diagram and practical-style questions.

Module 1: Measurement Tools and Accuracy

Short/viva: What is the principle of Vernier Calipers?

Diagram: Draw the setup/circuit.

Numerical: Use LC = 1 MSD - 1 VSD; sphere $V = \frac{4}{3} \pi r^3$; cylinder $V = \pi r^2 h$.

Practical: Mention precautions: Avoid parallax.; Do not overtighten jaws.; Use consistent units..

Short/viva: What is the principle of Screw Gauge?

Diagram: Draw the setup/circuit.

Numerical: Use $LC = \text{pitch} / \text{circular divisions}$; $\text{reading} = \text{PSR} + \text{CSR} \times LC \pm \text{zero correction}$.

Practical: Mention precautions: Use ratchet only.; Keep wire perpendicular.; Avoid compression..

Short/viva: What is the principle of Spherometer?

Diagram: Draw the setup/circuit.

Numerical: Use $R = \frac{l^2}{6h} + \frac{h}{2}$.

Practical: Mention precautions: All legs must touch surface.; Do not drag on optical surface.; Use correct sign..

Short/viva: What is the principle of Mercury Thermometer?

Diagram: Draw the setup/circuit.

Numerical: Use $K = C + 273$; $F = \frac{9C}{5} + 32$.

Practical: Mention precautions: Do not hold bulb.; Do not hit glass.; Read at eye level..

Module 2: Mechanics and Properties of Matter

Short/viva: What is the principle of Hooke Law?

Diagram: Draw the setup/circuit.

Numerical: Use $F = kx$; $k = F/x$.

Practical: Mention precautions: Do not cross elastic limit.; Wait until oscillation stops.; Load gently..

Short/viva: What is the principle of Stoke Law?

Diagram: Draw the setup/circuit.

Numerical: Use $\eta = \frac{2r^2(\rho_s - \rho_l)g}{9v}$.

Practical: Mention precautions: Avoid bubbles.; Sphere must fall centrally.; Use terminal region..

Short/viva: What is the principle of Parallelogram Law?

Diagram: Draw the setup/circuit.

Numerical: Use $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$.

Practical: Mention precautions: Pulleys friction-free.; Strings must meet at one point.; Use correct scale..

Short/viva: What is the principle of Moment Bar?

Diagram: Draw the setup/circuit.

Numerical: Use $W_1 d_1 = W_2 d_2$.

Practical: Mention precautions: Measure from pivot.; Keep knife edge sharp.; Avoid oscillation..

Short/viva: What is the principle of U-tube Apparatus?

Diagram: Draw the setup/circuit.

Numerical: Use $\rho_1 h_1 = \rho_2 h_2$.

Practical: Mention precautions: Avoid bubbles.; Read meniscus correctly.; Tube vertical..

Short/viva: What is the principle of Flywheel?

Diagram: Draw the setup/circuit.

Numerical: Use I from energy balance using mass, height, rotations and omega.

Practical: Mention precautions: No thread overlap.; Count rotations carefully.; Release without push..

Short/viva: What is the principle of Simple Pendulum?

Diagram: Draw the setup/circuit.

Numerical: Use $T = 2\pi \sqrt{l/g}$; $g = 4\pi^2 l / T^2$.

Practical: Mention precautions: Small amplitude.; Avoid elliptical motion.; Same reference point..

Short/viva: What is the principle of Resonance Column?

Diagram: Draw the setup/circuit.

Numerical: Use $v = 2f(l_2 - l_1)$.

Practical: Mention precautions: Do not touch fork.; Use same fork position.; Read water level at eye level..

Short/viva: What is the principle of Cantilever?

Diagram: Draw the setup/circuit.

Numerical: Use T depends on effective mass and stiffness.

Practical: Mention precautions: Clamp rigidly.; Small oscillation only.; Avoid air disturbance..

Module 3: Ray Optics Experiments

Short/viva: What is the principle of Convex Lens?

Diagram: Draw the setup/circuit.

Numerical: Use $1/f = 1/u + 1/v$; $P = 1/f$ metre.

Practical: Mention precautions: Same height.; Use sharp image.; Avoid parallax..

Short/viva: What is the principle of Snell Law?

Diagram: Draw the setup/circuit.

Numerical: Use $\mu = \sin i / \sin r$.

Practical: Mention precautions: Pins vertical.; Wide pin separation.; Draw normal accurately..

Module 4: Electrical and Semiconductor Characteristics

Short/viva: What is the principle of Ohm Law?

Diagram: Draw the setup/circuit.

Numerical: Use $V = IR$; $R = V/I$.

Practical: Mention precautions: Check polarity.; Avoid heating.; Use proper range..

Short/viva: What is the principle of Laws of Resistances?

Diagram: Draw the setup/circuit.

Numerical: Use $R_s = R_1 + R_2$; $1/R_p = 1/R_1 + 1/R_2$.

Practical: Mention precautions: Correct wiring.; No short circuit.; Proper meter range..

Short/viva: What is the principle of Diode V-I Characteristics?

Diagram: Draw the setup/circuit.

Numerical: Use Ge knee about 0.3 V; Si knee about 0.7 V.

Practical: Mention precautions: Use current limiting resistor.; Do not exceed rating.; Check polarity..

Short/viva: What is the principle of Potentiometer?

Diagram: Draw the setup/circuit.

Numerical: Use $E_1/E_2 = l_1/l_2$; $r = R(l_1 - l_2)/l_2$.

Practical: Mention precautions: Do not drag jockey.; Driver emf must be higher.; Check polarity..

Short/viva: What is the principle of Meter Bridge?

Diagram: Draw the setup/circuit.

Numerical: Use $X/R = l/(100-l)$; $\rho = RA/L$.

Practical: Mention precautions: Do not press jockey hard.; Balance near middle.; Remove key when idle..

Objective Practice Answer Key

1. Least count means smallest measurable value. 2. Ohm law graph slope gives resistance. 3. Meter bridge is based on Wheatstone bridge. 4. Snell law gives refractive index. 5. Diode knee voltage is read from forward V-I curve.

Fresh Model Question Paper

Prepared for practice from the supplied syllabus. It does not copy official questions.

Time: 3 Hours **Maximum marks:** 75 **Pattern:** Fresh practice paper based on lab syllabus.

Part A - Answer all, 2 marks each

1. Define least count and zero error.

2. State Hooke law.

3. What is terminal velocity?

4. State Snell law.

5. What is diode knee voltage?

6. Write balance condition in meter bridge.

7. Mention two thermometer precautions.

8. Why is a current-limiting resistor used with diode?

Part B - Answer any six, 5 marks each

1. Explain screw gauge diameter measurement.

2. Describe parallelogram law experiment.

3. Explain simple pendulum method to find g.

4. Describe resonance column method.

5. Explain u-v method for convex lens.

6. Verify Ohm law using V-I graph.

7. Draw diode V-I characteristics and find knee voltage.

8. Find specific resistance using meter bridge.

Part C - Answer any four, 10 marks each

1. With diagram, explain Vernier calipers experiment for volume measurement.

2. Explain Stoke law viscosity experiment with formula and precautions.

3. Explain flywheel experiment and energy idea.

4. Verify Snell law using glass slab/prism with ray diagram.

5. Verify laws of series and parallel resistances with circuits.

6. Explain potentiometer method for emf comparison/internal resistance.

Complete Model Answers

Answer guide with expected points, formula/diagram hints, precautions and common mistakes.

General answer format

Every model answer should contain definition/principle, labelled diagram or circuit, formula with variables, procedure, observation table, calculation, result with unit, precautions and common mistakes.

Answer 1: Vernier calipers: define least count, zero correction, jaws, $MSR + VSR \times LC$, volume formula and precautions.

Answer 2: Screw gauge: pitch, LC, $PSR + CSR \times LC$, ratchet use and zero correction.

Answer 3: Hooke law: F proportional to x , plot graph, slope gives k , stay within elastic limit.

Answer 4: Stoke law: terminal velocity, eta formula, central fall, avoid bubbles, use terminal region.

Answer 5: Optics: draw normal/rays, use lens equation or Snell ratio, avoid parallax and thick lines.

Answer 6: Electrical: show correct ammeter/voltmeter connection, polarity, current limiting, graph slope and loose-contact troubleshooting.

Answer 7: Meter bridge/potentiometer: null method, balance condition, jockey precautions and connection polarity.

References

- Text Book of Physics for Class XI & XII (Part-I, Part-II), N.C.E.R.T., Delhi.
- Comprehensive Practical Physics, Vol. I & II, J. N. Jaiswal, Laxmi Publications (P) Ltd., New Delhi.
- Practical Physics by C. L. Arora, S. Chand & Company Ltd.

Online resources: Supplied source mentions e-books/e-tools/learning physics software/YouTube videos/websites etc. without specific URLs. Official references with exact URLs were not specified in supplied source; use trusted institutional resources as supplementary learning links.